Equations of Planes: Vector Equations

Questions 1–12

Question 1. Find a vector equation of the planes that pass through these points:

a) $(1, 0, 1)$, $(2, 1, 0)$ and $(1, 2, 2)$

b) $(-1, 2, -3)$, $(3, 2, 3)$ and $(-1, -1, 2)$

c) $(0, 0, 0)$, $(3, 4, 1)$ and $(-9, -3, 7)$

Question 2. Determine whether each of these points lies on the plane with equation

$$\mathbf{r} = \begin{pmatrix} 3 \\ 2 \\ -5 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ 2 \\ 1 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ 3 \\ -1 \end{pmatrix} :$$

a) $(9, 18, -7)$

b) $(24, 23, -17)$

c) $(19, 5, -22)$

Question 3. It is given that $A(2, -3, 5)$, $B(-6, 2, 3)$ and $C(34, -23, 13)$ relative to a fixed origin O.

a) Find λ and μ such that $\overrightarrow{OC} = \lambda\overrightarrow{OA} + \mu\overrightarrow{OB}$.

b) Explain what your answer to part (a) tells you about the points O, A, B and C.

Question 4. The plane Π has equation

$$\mathbf{r} = \begin{pmatrix} -2 \\ 1 \\ 4 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ -9 \\ 8 \end{pmatrix} + \mu \begin{pmatrix} 3 \\ 7 \\ -14 \end{pmatrix}.$$

a) Show that the vector $35\mathbf{i} + 33\mathbf{j} + 24\mathbf{k}$ is a normal vector to the plane.

b) Hence find the equation of the plane Π in cartesian form, $ax + by + cz = d$, where a, b, c and d are integers.

Question 5. Determine whether the line with equation $\mathbf{r} = 2\mathbf{i} + \lambda(-\mathbf{i} + 2\mathbf{j} - 3\mathbf{k})$ lies in the plane with equation $\mathbf{r} = \mathbf{j} + \alpha(\mathbf{i} - \mathbf{k}) + \beta(\mathbf{i} - 2\mathbf{j} + 3\mathbf{k})$.

Question 6. Find the acute angle between the following pairs of planes:

a) $\Pi_1 : \mathbf{r} = \begin{pmatrix} 1 \\ 0 \\ -4 \end{pmatrix} + \alpha \begin{pmatrix} 2 \\ 1 \\ 4 \end{pmatrix} + \beta \begin{pmatrix} 3 \\ 1 \\ 0 \end{pmatrix}$, $\Pi_2 : \mathbf{r} = \begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix} + \gamma \begin{pmatrix} 1 \\ 3 \\ -2 \end{pmatrix} + \delta \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix}$

b) $\Pi_1 : \mathbf{r} = \begin{pmatrix} 6 \\ 2 \\ -5 \end{pmatrix} + \alpha \begin{pmatrix} 4 \\ 0 \\ 5 \end{pmatrix} + \beta \begin{pmatrix} 3 \\ -2 \\ 0 \end{pmatrix}$, $\Pi_2 : \mathbf{r} = \begin{pmatrix} 4 \\ 3 \\ 0 \end{pmatrix} + \gamma \begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix} + \delta \begin{pmatrix} 3 \\ -2 \\ 3 \end{pmatrix}$

Question 7. Write the equation of each of these planes in cartesian form, $ax + by + cz = d$, where a, b, c and d are integers:

a) $\mathbf{r} = \begin{pmatrix}0\\1\\0\end{pmatrix} + \lambda\begin{pmatrix}1\\0\\0\end{pmatrix} + \mu\begin{pmatrix}0\\1\\1\end{pmatrix}$

b) $\mathbf{r} = \begin{pmatrix}1\\2\\5\end{pmatrix} + \lambda\begin{pmatrix}2\\1\\1\end{pmatrix} + \mu\begin{pmatrix}4\\3\\1\end{pmatrix}$

c) $\mathbf{r} = \begin{pmatrix}4\\2\\-5\end{pmatrix} + \lambda\begin{pmatrix}3\\2\\0\end{pmatrix} + \mu\begin{pmatrix}0\\-5\\6\end{pmatrix}$

d) $\mathbf{r} = \begin{pmatrix}-2\\-8\\10\end{pmatrix} + \lambda\begin{pmatrix}-2\\4\\-7\end{pmatrix} + \mu\begin{pmatrix}5\\0\\-4\end{pmatrix}$

Question 8. Write the equation of each of these planes in cartesian form, $ax + by + cz = d$, where a, b, c and d are integers:

a) $\mathbf{r} = \mathbf{i} - 10\mathbf{j} - 2\mathbf{k} + \lambda(2\mathbf{i} + \mathbf{j} + 5\mathbf{k}) + \mu(-3\mathbf{i} - 2\mathbf{j} + 9\mathbf{k})$

b) $\mathbf{r} = \lambda(2\mathbf{i} + 3\mathbf{j} - 15\mathbf{k}) + \mu(8\mathbf{i} + 7\mathbf{k})$

Question 9. Write the equation of each of these planes in the form $\mathbf{r} \cdot \mathbf{n} = p$:

a) $\mathbf{r} = \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 0 \\ 1 \\ 3 \end{pmatrix} + \mu \begin{pmatrix} -2 \\ 3 \\ 2 \end{pmatrix}$

b) $\mathbf{r} = -8\mathbf{j} + 16\mathbf{k} + \lambda(20\mathbf{i} - 8\mathbf{j} + 5\mathbf{k}) + \mu(15\mathbf{i} + 8\mathbf{j})$

Question 10. Three planes, Π_1, Π_2 and Π_3, have equations:

$$\Pi_1 : 2x - 3y + 8z = -2, \qquad \Pi_2 : \mathbf{r} \cdot \begin{pmatrix} 4 \\ 1 \\ 2 \end{pmatrix} = 3, \qquad \Pi_3 : \mathbf{r} = \begin{pmatrix} 3 \\ -5 \\ 6 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 6 \\ -3 \end{pmatrix} + \mu \begin{pmatrix} 5 \\ 4 \\ -1 \end{pmatrix}.$$

a) Find the acute angle between Π_1 and Π_2.

b) Find the acute angle between Π_1 and Π_3.

Question 11. A plane contains the points $A(0, 2, 6)$, $B(-3, 1, 0)$ and $C(3, 0, -4)$. Find the equation of the plane in the form $ax + by + cz + d = 0$, where a, b, c and d are integers.

Question 12. A plane contains the points $A(4, -5, 1)$, $B(2, 3, 12)$ and $C(6, -2, 10)$. Find the equation of the plane in the form $(\mathbf{r} - \mathbf{a}) \cdot \mathbf{n} = 0$.